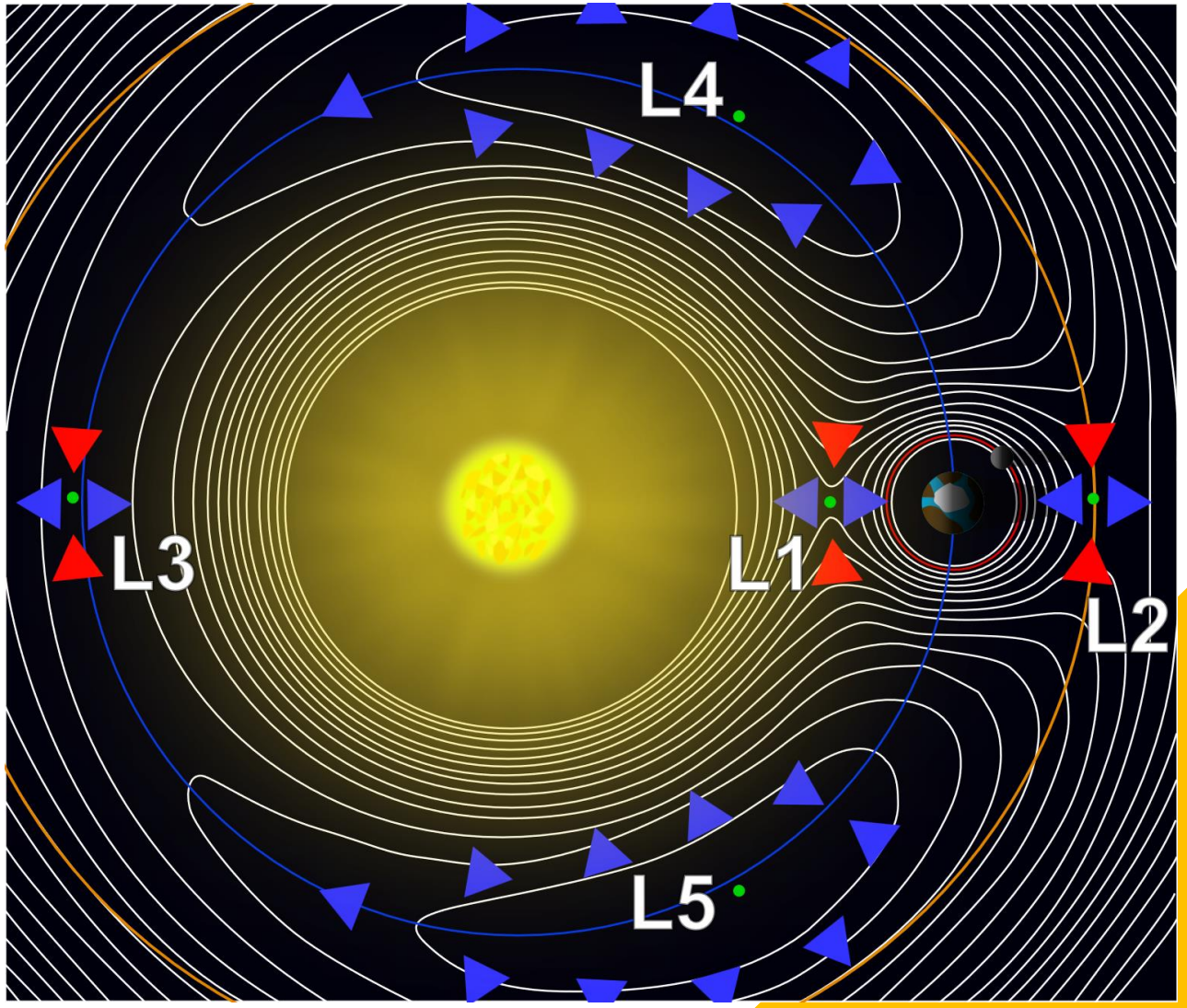
PHYSICS I
WEEK 11

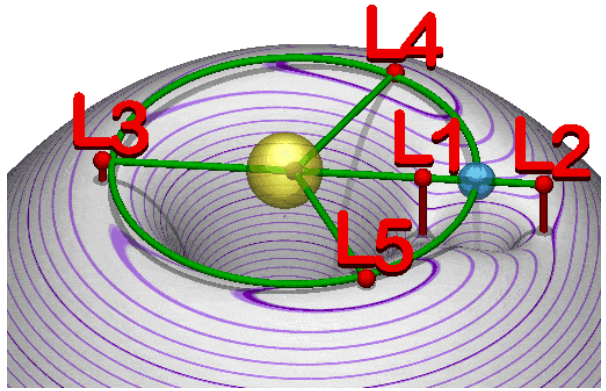
Siddharth Dhomkar
Department of Physics
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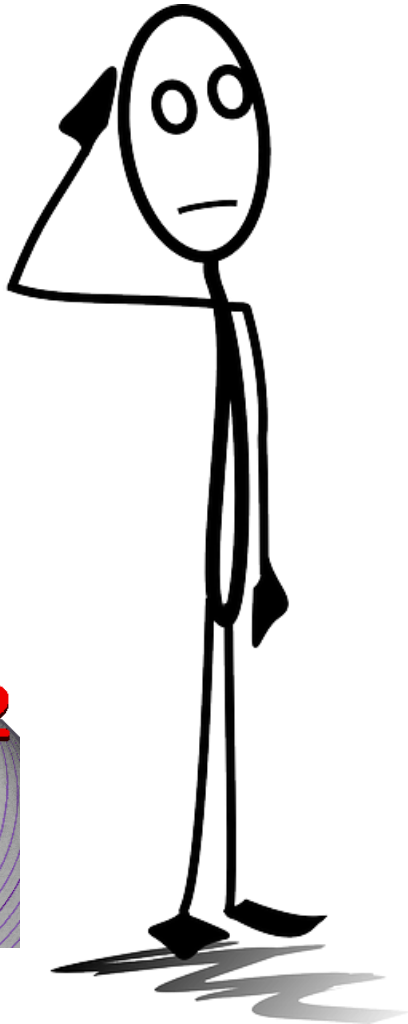


WEEK 11: OUTLINE

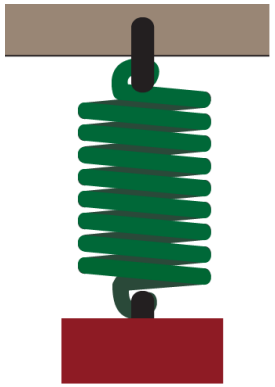
- Review: Lagrangian Dynamics
- Motion under Central Forces:
  - Lagrangian in plane polar coordinates:

    Morin 7.1, 7.2
  - Gravity: Morin 7.3, 7.4





REVIEW: EULER-LAGRANGE EQUATIONS



Newton' law

$$F = ma$$

$$m\ddot{x} = -kx$$

$$L = T - U$$

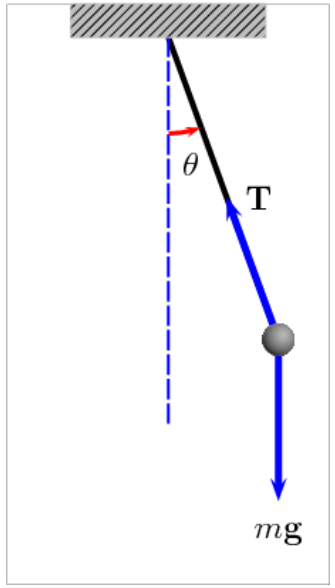
$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = \frac{\partial L}{\partial x}$$

Euler-Lagrange equations

REVIEW: EULER-LAGRANGE EQUATIONS

Simple pendulum

$$x = l\sin\theta$$
$$y = l\cos\theta$$



$$T = \frac{1}{2}mv^2 \qquad U = -mgl\cos\theta$$

$$L = T - U$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = \frac{\partial L}{\partial x}$$

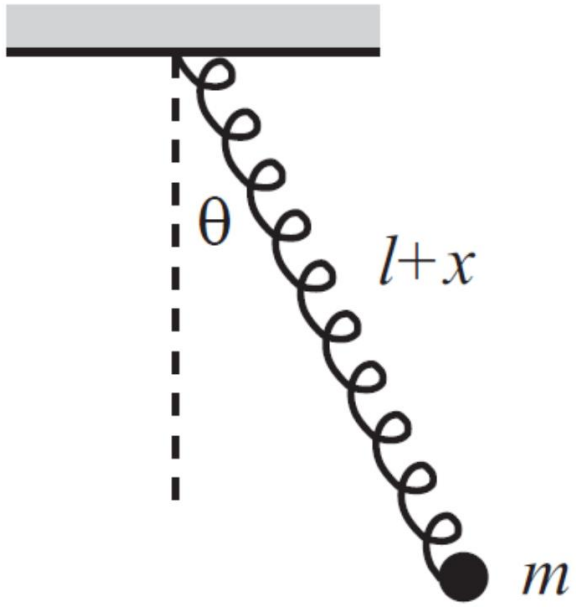
Euler-Lagrange equations

Constraint equation

$$r = l$$

$$\ddot{\theta} = -\frac{g}{l}\sin\theta$$

REVIEW: EULER-LAGRANGE EQUATIONS



$$T = \frac{1}{2}m\left(\dot{x}^2 + (\ell + x)^2\dot{\theta}^2\right) \qquad V(x,\theta) = -mg(\ell + x)\cos\theta + \frac{1}{2}kx^2$$

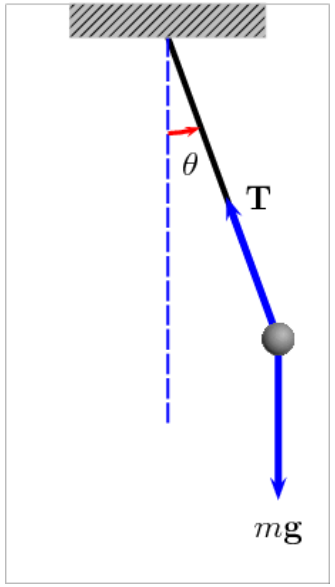
$$L \equiv T - V = \frac{1}{2}m\left(\dot{x}^2 + (\ell + x)^2\dot{\theta}^2\right) + mg(\ell + x)\cos\theta - \frac{1}{2}kx^2$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = \frac{\partial L}{\partial x} \quad \Longrightarrow \quad m\ddot{x} = m(\ell + x)\dot{\theta}^2 + mg\cos\theta - kx$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = \frac{\partial L}{\partial \theta} \quad \Longrightarrow \quad m(\ell + x)\ddot{\theta} + 2m\dot{x}\dot{\theta} = -mg\sin\theta$$

REVIEW: GENERALIZED COORDINATES

Simple pendulum



$$r = l$$

$$q = \theta$$

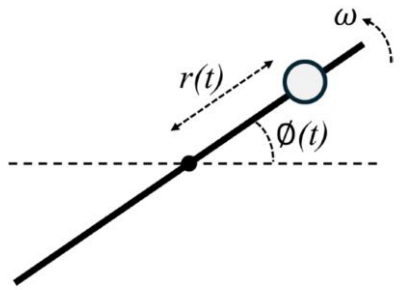
Generalized coordinates

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_k}\right) = \frac{\partial L}{\partial q_k}$$

- If Cartesian coordinates are used, then ${}^{\partial L}/_{\partial \dot{x}_k}$ is simply the momentum, $m\dot{x}_k$.
- ${}^{\partial L}/_{\partial \dot{q}_k}$ is called the **generalized momentum**.

REVIEW: TUTORIAL 7

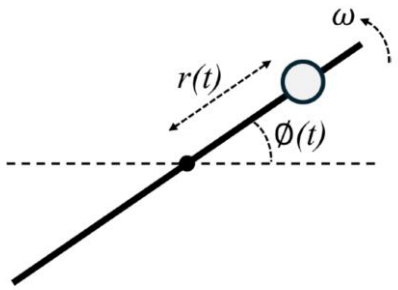


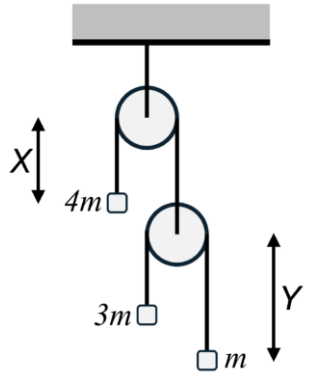
$$\mathcal{L} = T - V$$
$$= \frac{1}{2}m\left(\dot{r}^2 + \omega^2 r^2\right)$$

$$\frac{d^2r}{dt^2} - \omega^2 r = 0$$

$$r(t) = Ae^{\omega t} + Be^{-\omega t}$$

REVIEW: TUTORIAL 7





$$\mathcal{L} = T - V$$
$$= \frac{1}{2}m\left(\dot{r}^2 + \omega^2 r^2\right)$$

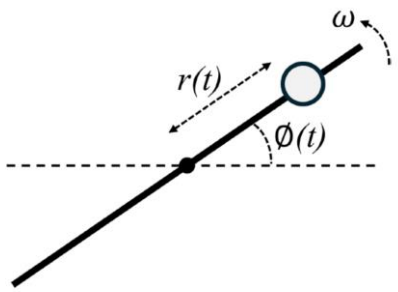
$$\mathcal{L} = m\left(4\dot{x}^2 + 2\dot{y}^2 + 2\dot{x}\dot{y}\right)$$
$$- 2mgy$$

$$\frac{d^2r}{dt^2} - \omega^2 r = 0$$

$$\ddot{x} = g/7$$

$$r(t) = Ae^{\omega t} + Be^{-\omega t}$$

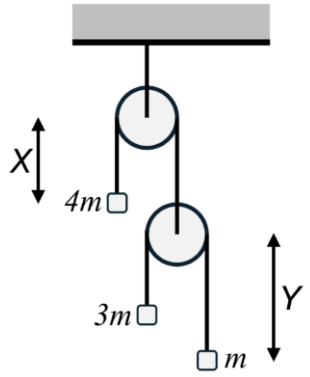
REVIEW: TUTORIAL 7



$$\mathcal{L} = T - V$$
$$= \frac{1}{2}m\left(\dot{r}^2 + \omega^2 r^2\right)$$

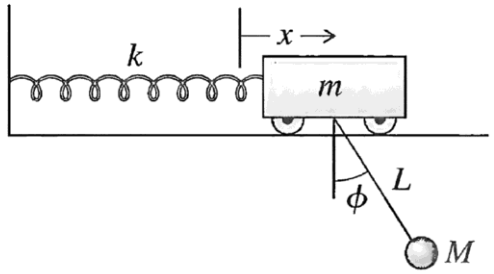
$$\frac{d^2 r}{dt^2} - \omega^2 r = 0$$

$$r(t) = Ae^{\omega t} + Be^{-\omega t}$$



$$\mathcal{L} = m\left(4\dot{x}^2 + 2\dot{y}^2 + 2\dot{x}\dot{y}\right) - 2mgy$$

$$\ddot{x} = g/7$$



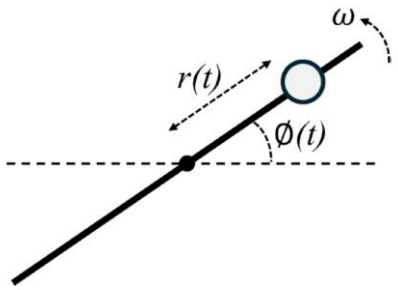
$$x_M = x + L\sin\varphi,$$
$$y_M = L\cos\varphi$$

$$\mathcal{L} = \frac{1}{2}(M+m)\dot{x}^2 + \frac{1}{2}ML^2\dot{\varphi}^2 + ML\cos\varphi\dot{x}\dot{\varphi} - \frac{1}{2}kx^2 + MgL\cos\varphi$$

$$(M+m)\ddot{x} + ML\cos\varphi\ddot{\varphi} - ML\sin\varphi(\dot{\varphi})^2 + kx = 0$$

$$L\ddot{\varphi} + \cos\varphi\ddot{x} + g\sin\varphi = 0$$

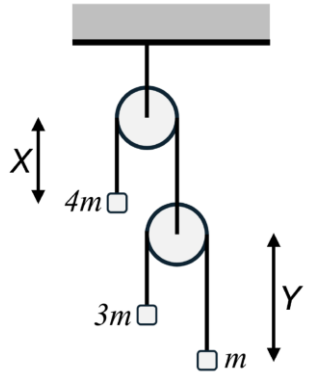
REVIEW: TUTORIAL 7



$$\mathcal{L} = T - V$$
$$= \frac{1}{2}m\left(\dot{r}^2 + \omega^2 r^2\right)$$

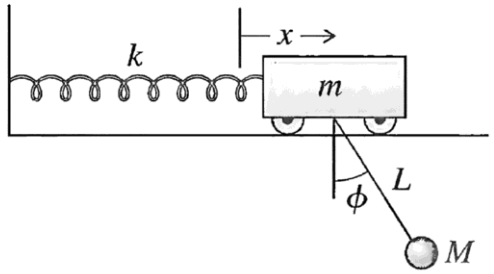
$$\frac{d^2r}{dt^2} - \omega^2 r = 0$$

$$r(t) = Ae^{\omega t} + Be^{-\omega t}$$



$$\mathcal{L} = m\left(4\dot{x}^2 + 2\dot{y}^2 + 2\dot{x}\dot{y}\right) - 2mgy$$

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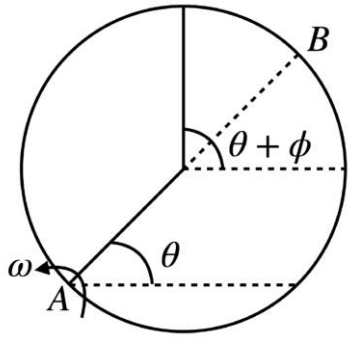


$$x_M = x + L\sin\varphi,$$
$$y_M = L\cos\varphi$$

$$\mathcal{L} = \frac{1}{2}(M+m)\dot{x}^2 + \frac{1}{2}ML^2\dot{\varphi}^2 + ML\cos\varphi\dot{x}\dot{\varphi} - \frac{1}{2}kx^2 + MgL\cos\varphi$$

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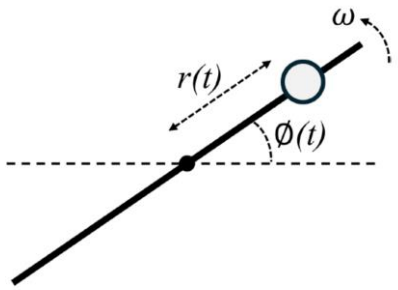
$$x = R\cos\theta + R\cos(\theta+\varphi)$$
$$y = R\sin\theta + R\sin(\theta+\varphi)$$

$$\mathcal{L} = \frac{1}{2}mR^2\left[\omega^2 + (\omega+\dot{\varphi})^2 + 2\omega(\omega+\dot{\varphi})\cos\varphi\right]$$

$$\ddot{\varphi} + \omega^2\sin\varphi = 0$$

simple pendulum

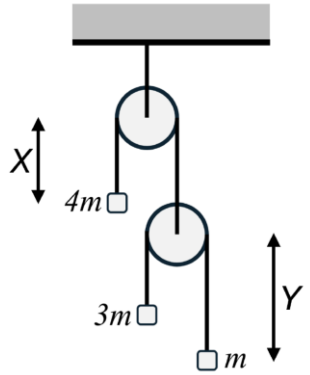
REVIEW: TUTORIAL 7



$$\mathcal{L} = T - V = \frac{1}{2}m\left(\dot{r}^2 + \omega^2 r^2\right)$$

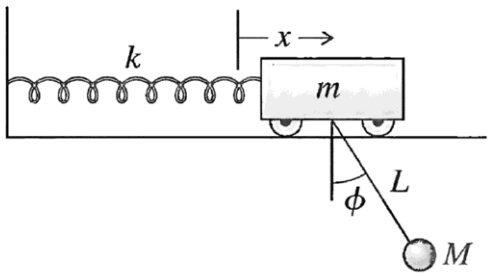
$$\frac{d^2 r}{dt^2} - \omega^2 r = 0$$

$$r(t) = Ae^{\omega t} + Be^{-\omega t}$$



$$\mathcal{L} = m\left(4\dot{x}^2 + 2\dot{y}^2 + 2\dot{x}\dot{y}\right) - 2mgy$$

$$\ddot{x} = g/7$$

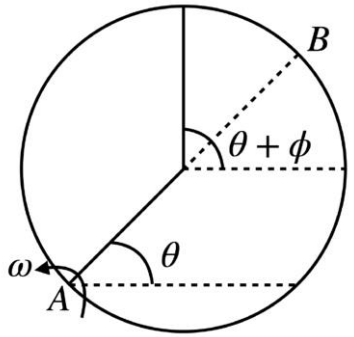


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$$(M+m)\ddot{x} + ML\cos\varphi\ddot{\varphi} - ML\sin\varphi(\dot{\varphi})^2 + kx = 0$$

$$L\ddot{\varphi} + \cos\varphi\ddot{x} + g\sin\varphi = 0$$

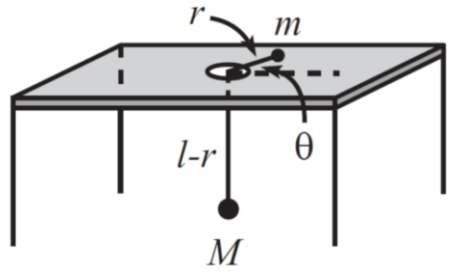


$$x = R\cos\theta + R\cos(\theta+\varphi)$$
$$y = R\sin\theta + R\sin(\theta+\varphi)$$

$$\mathcal{L} = \frac{1}{2}mR^2\left[\omega^2 + (\omega+\dot{\varphi})^2 + 2\omega(\omega+\dot{\varphi})\cos\varphi\right]$$

$$\ddot{\varphi} + \omega^2\sin\varphi = 0$$

simple pendulum



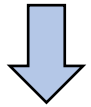
$$\mathcal{L} = \frac{1}{2}M\dot{r}^2 + m(\dot{r}^2 + r^2\dot{\theta}^2) + Mg(l-r)$$

$$(M+m)\ddot{r} = \frac{L^2}{mr^3} - Mg$$

REVIEW: CONSERVATION LAWS

Cyclic coordinates

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_k}\right) = \frac{\partial L}{\partial q_k} = 0$$



$$\frac{\partial L}{\partial \dot{q}_k} = C$$

Conserved Quantity

Example 1: Consider a ball thrown through the air.

$$L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgz$$

Conserved Quantity: Linear momentum

Example 2: $V(r, \theta)$ in spherical coordinates.

$$L = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2 sin^2\theta\dot{\emptyset}^2\right) - V(r, \theta)$$

Conserved Quantity: Angular momentum

Example 3: $V(r)$ in cylindrical coordinates.

$$L = \frac{1}{2}m\left(\dot{\rho}^2 + \rho^2\dot{\emptyset}^2 + \dot{z}^2\right) - V(r)$$

Conserved Quantity: Linear & angular momentum

REVIEW: SPHERICAL COORDINATE SYSTEM

Position vector $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$

$$\Rightarrow \boxed{\vec{r} = r\sin\theta\cos\phi\,\hat{x} + r\sin\theta\sin\phi\,\hat{y} + r\cos\theta\,\hat{z}}$$

$$\hat{r} = \frac{\partial\vec{r}/\partial r}{\left|\partial\vec{r}/\partial r\right|} = \sin\theta\cos\phi\,\hat{x} + \sin\theta\sin\phi\,\hat{y} + \cos\theta\,\hat{z}$$

$$\hat{\theta} = \frac{\partial\vec{r}/\partial\theta}{\left|\partial\vec{r}/\partial\theta\right|} = \cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z}$$

$$\hat{\phi} = \frac{\partial\vec{r}/\partial\phi}{\left|\partial\vec{r}/\partial\phi\right|} = -\sin\phi\,\hat{x} + \cos\phi\,\hat{y}$$

Velocity

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dr}{dt}\hat{r} + r\frac{d\hat{r}}{dt}$$

$$\frac{d\hat{r}}{dt} = \left(\cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z}\right)\dot{\theta} + \sin\theta\left(-\sin\phi\,\hat{x} + \cos\phi\,\hat{y}\right)\dot{\phi}$$

$$= \dot{\theta}\hat{\theta} + \sin\theta\,\dot{\phi}\,\hat{\phi}$$

$$\Rightarrow \boxed{\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + r\sin\theta\,\dot{\phi}\,\hat{\phi}}$$

REVIEW: SPHERICAL COORDINATE SYSTEM

Position vector $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$

$$\Rightarrow \boxed{\vec{r} = r\sin\theta\cos\phi\,\hat{x} + r\sin\theta\sin\phi\,\hat{y} + r\cos\theta\,\hat{z}}$$

$$\hat{r} = \frac{\partial\vec{r}/\partial r}{\left|\partial\vec{r}/\partial r\right|} = \sin\theta\cos\phi\,\hat{x} + \sin\theta\sin\phi\,\hat{y} + \cos\theta\,\hat{z}$$

$$\hat{\theta} = \frac{\partial\vec{r}/\partial\theta}{\left|\partial\vec{r}/\partial\theta\right|} = \cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z}$$

$$\hat{\phi} = \frac{\partial\vec{r}/\partial\phi}{\left|\partial\vec{r}/\partial\phi\right|} = -\sin\phi\,\hat{x} + \cos\phi\,\hat{y}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dr}{dt}\hat{r} + r\frac{d\hat{r}}{dt}$$

$$\frac{d\hat{r}}{dt} = \left(\cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z}\right)\dot{\theta} + \sin\theta\left(-\sin\phi\,\hat{x} + \cos\phi\,\hat{y}\right)\dot{\phi}$$

$$= \dot{\theta}\hat{\theta} + \sin\theta\,\dot{\phi}\hat{\phi}$$

$$\Rightarrow \boxed{\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + r\sin\theta\,\dot{\phi}\hat{\phi}}$$

Velocity, acceleration, kinetic energy

$$\boldsymbol{a} = \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = \frac{\mathrm{d}\left(\dot{r}\,\hat{\boldsymbol{r}}\right)}{\mathrm{d}t} + \frac{\mathrm{d}\left(r\,\dot{\theta}\,\hat{\boldsymbol{\theta}}\right)}{\mathrm{d}t} + \frac{\mathrm{d}\left(r\sin\theta\,\dot{\phi}\,\hat{\boldsymbol{\phi}}\right)}{\mathrm{d}t}$$

$$T = \frac{m}{2}\,\boldsymbol{v}\cdot\boldsymbol{v} = \frac{m}{2}\left[\dot{r}^2 + \left(r\,\dot{\theta}\right)^2 + \left(r\sin\theta\,\dot{\phi}\right)^2\right]$$

REVIEW: SPHERICAL COORDINATE SYSTEM

Position vector $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$

$$\Rightarrow \boxed{\vec{r} = r\sin\theta\cos\phi\,\hat{x} + r\sin\theta\sin\phi\,\hat{y} + r\cos\theta\,\hat{z}}$$

$$\hat{r} = \frac{\partial\vec{r}/\partial r}{\left|\partial\vec{r}/\partial r\right|} = \sin\theta\cos\phi\,\hat{x} + \sin\theta\sin\phi\,\hat{y} + \cos\theta\,\hat{z}$$

$$\hat{\theta} = \frac{\partial\vec{r}/\partial\theta}{\left|\partial\vec{r}/\partial\theta\right|} = \cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z}$$

$$\hat{\phi} = \frac{\partial\vec{r}/\partial\phi}{\left|\partial\vec{r}/\partial\phi\right|} = -\sin\phi\,\hat{x} + \cos\phi\,\hat{y}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dr}{dt}\hat{r} + r\frac{d\hat{r}}{dt}$$

$$\frac{d\hat{r}}{dt} = \left(\cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z}\right)\dot{\theta} + \sin\theta\left(-\sin\phi\,\hat{x} + \cos\phi\,\hat{y}\right)\dot{\phi}$$

$$= \dot{\theta}\hat{\theta} + \sin\theta\,\dot{\phi}\,\hat{\phi}$$

$$\Rightarrow \boxed{\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + r\sin\theta\,\dot{\phi}\,\hat{\phi}}$$

Velocity, acceleration, kinetic energy, and angular momentum

$$\boldsymbol{a} = \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = \frac{\mathrm{d}\left(\dot{r}\,\hat{\boldsymbol{r}}\right)}{\mathrm{d}t} + \frac{\mathrm{d}\left(r\,\dot{\theta}\,\hat{\boldsymbol{\theta}}\right)}{\mathrm{d}t} + \frac{\mathrm{d}\left(r\sin\theta\,\dot{\phi}\,\hat{\boldsymbol{\phi}}\right)}{\mathrm{d}t}$$

$$T = \frac{m}{2}\,\boldsymbol{v}\cdot\boldsymbol{v} = \frac{m}{2}\left[\dot{r}^2 + \left(r\,\dot{\theta}\right)^2 + \left(r\sin\theta\,\dot{\phi}\right)^2\right]$$

$$\vec{L} = (\vec{r}\times\vec{p}) = m(\vec{r}\times\vec{v})$$

REVIEW: SPHERICAL COORDINATE SYSTEM

Position vector $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$

$$\Rightarrow \boxed{\vec{r} = r\sin\theta\cos\phi\,\hat{x} + r\sin\theta\sin\phi\,\hat{y} + r\cos\theta\,\hat{z}}$$

$$\hat{r} = \frac{\partial\vec{r}/\partial r}{\left|\partial\vec{r}/\partial r\right|} = \sin\theta\cos\phi\,\hat{x} + \sin\theta\sin\phi\,\hat{y} + \cos\theta\,\hat{z}$$

$$\hat{\theta} = \frac{\partial\vec{r}/\partial\theta}{\left|\partial\vec{r}/\partial\theta\right|} = \cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z}$$

$$\hat{\phi} = \frac{\partial\vec{r}/\partial\phi}{\left|\partial\vec{r}/\partial\phi\right|} = -\sin\phi\,\hat{x} + \cos\phi\,\hat{y}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dr}{dt}\hat{r} + r\frac{d\hat{r}}{dt}$$

$$\frac{d\hat{r}}{dt} = \left(\cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z}\right)\dot{\theta} + \sin\theta\left(-\sin\phi\,\hat{x} + \cos\phi\,\hat{y}\right)\dot{\phi}$$

$$= \dot{\theta}\hat{\theta} + \sin\theta\,\dot{\phi}\,\hat{\phi}$$

$$\Rightarrow \boxed{\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} + r\sin\theta\,\dot{\phi}\,\hat{\phi}}$$

Velocity, acceleration, kinetic energy, and angular momentum

$$\boldsymbol{a} = \frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = \frac{\mathrm{d}\left(\dot{r}\,\hat{\boldsymbol{r}}\right)}{\mathrm{d}t} + \frac{\mathrm{d}\left(r\,\dot{\theta}\,\hat{\boldsymbol{\theta}}\right)}{\mathrm{d}t} + \frac{\mathrm{d}\left(r\sin\theta\,\dot{\phi}\,\hat{\boldsymbol{\phi}}\right)}{\mathrm{d}t}$$

$$T = \frac{m}{2}\,\boldsymbol{v}\cdot\boldsymbol{v} = \frac{m}{2}\left[\dot{r}^2 + \left(r\,\dot{\theta}\right)^2 + \left(r\sin\theta\,\dot{\phi}\right)^2\right]$$

$$\vec{L} = (\vec{r}\times\vec{p}) = m(\vec{r}\times\vec{v})$$

$$\boldsymbol{L} = -mr^2\sin\theta\dot{\phi}\widehat{\boldsymbol{\theta}} + mr^2\dot{\theta}\widehat{\phi}$$

REVIEW: SPHERICAL COORDINATE SYSTEM

Position vector $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$

$$\Rightarrow \boxed{\vec{r} = r\sin\theta\cos\phi\,\hat{x} + r\sin\theta\sin\phi\,\hat{y} + r\cos\theta\,\hat{z}}$$

$$\hat{r} = \frac{\partial\vec{r}/\partial r}{\left|\partial\vec{r}/\partial r\right|} = \sin\theta\cos\phi\,\hat{x} + \sin\theta\sin\phi\,\hat{y} + \cos\theta\,\hat{z}$$

$$\hat{\theta} = \frac{\partial\vec{r}/\partial\theta}{\left|\partial\vec{r}/\partial\theta\right|} = \cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z}$$

$$\hat{\phi} = \frac{\partial\vec{r}/\partial\phi}{\left|\partial\vec{r}/\partial\phi\right|} = -\sin\phi\,\hat{x} + \cos\phi\,\hat{y}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dr}{dt}\hat{r} + r\frac{d\hat{r}}{dt}$$

$$\frac{d\hat{r}}{dt} = \left(\cos\theta\cos\phi\,\hat{x} + \cos\theta\sin\phi\,\hat{y} - \sin\theta\,\hat{z}\right)\dot{\theta} + \sin\theta\left(-\sin\phi\,\hat{x} + \cos\phi\,\hat{y}\right)\dot{\phi}$$

$$= \dot{\theta}\hat{\theta} + \sin\theta\,\dot{\phi}\,\hat{\phi}$$

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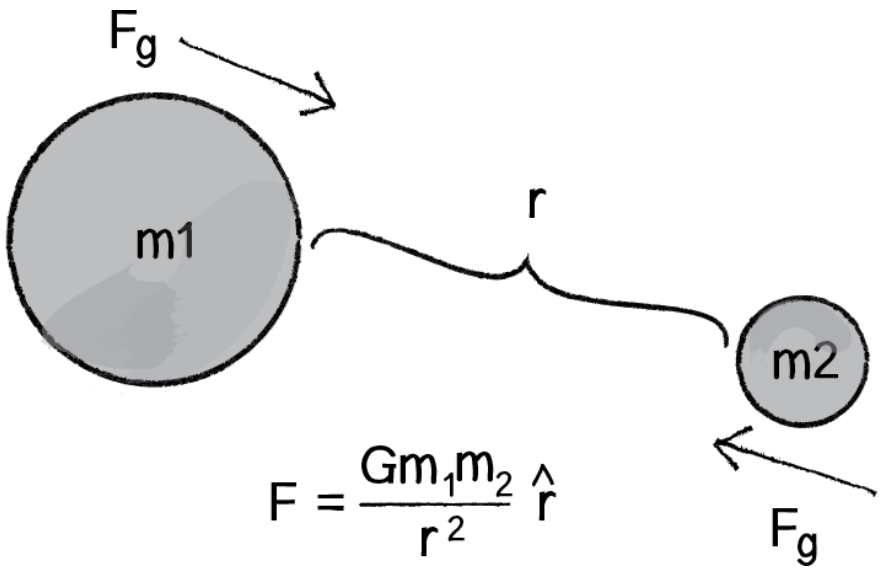
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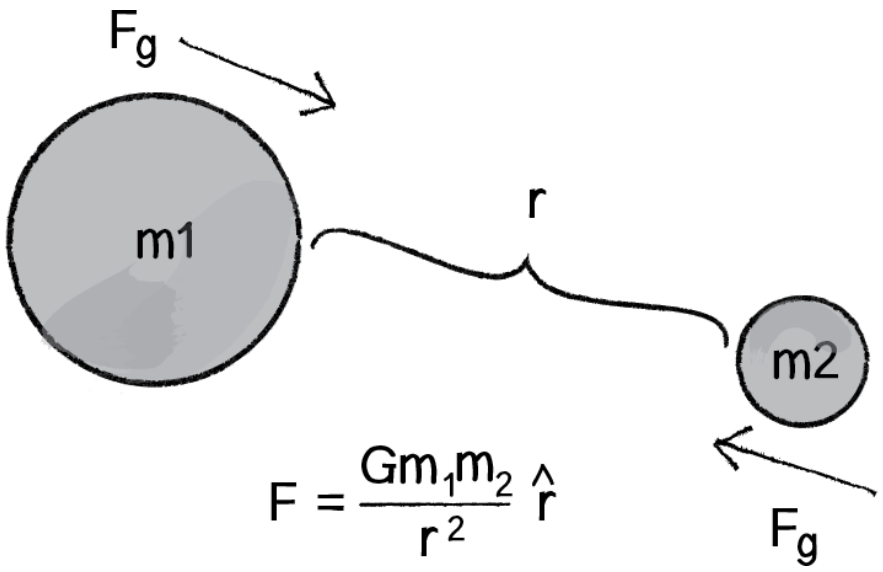
CENTRAL FORCE



Electrostatic and gravitational forces $V(r) \propto \frac{1}{r}$

Spring force $V(r) \propto (r - l)^2$

CENTRAL FORCE



Electrostatic and gravitational forces $V(r) \propto \frac{1}{r}$

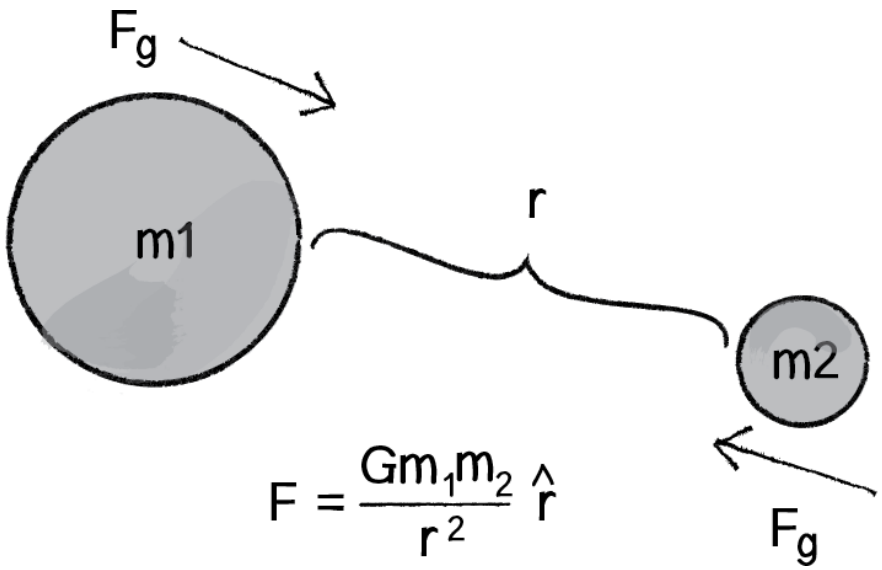
Spring force $V(r) \propto (r-l)^2$

$$\boldsymbol{F \propto r}$$

If a particle is subject to a central force only, then its angular momentum is conserved.

$$\begin{aligned}\frac{d\mathbf{L}}{dt} &= \frac{d}{dt}(\mathbf{r}\times\mathbf{p}) \\ &= \frac{d\mathbf{r}}{dt}\times\mathbf{p} + \mathbf{r}\times\frac{d\mathbf{p}}{dt} \\ &= \mathbf{v}\times(m\mathbf{v}) + \mathbf{r}\times\mathbf{F} \\ &= 0,\end{aligned}$$

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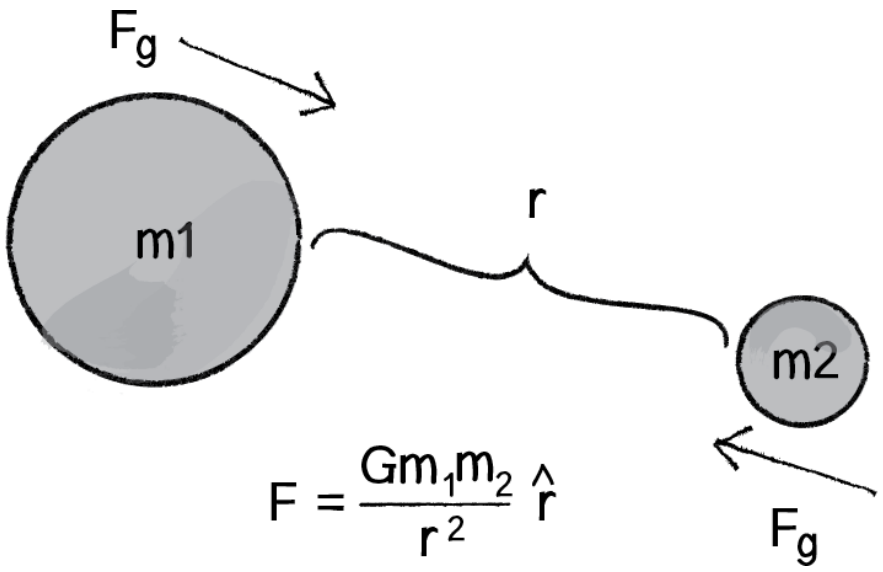
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If a particle is subject to a central force only, then its motion takes place in a plane.

$$\mathbf{r}\times\mathbf{v} = \mathbf{n}_0 \text{ for all } t$$

CENTRAL FORCE



$$\mathbf{F}(\mathbf{r}) = -\nabla V(r) = -\frac{dV}{dr}\hat{\mathbf{r}}$$

$$\mathcal{L} = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right) - V(r)$$

Electrostatic and gravitational forces $V(r) \propto \frac{1}{r}$

Spring force $V(r) \propto (r-l)^2$

Equations of motion

$$m\ddot{r} = mr\dot{\theta}^2 - V'(r), \qquad \frac{d}{dt}(mr^2\dot{\theta}) = 0.$$

CENTRAL FORCE: THE SOLUTION

$$\frac{1}{2}m\dot{r}^2 + \left(\frac{L^2}{2mr^2} + V(r)\right) = E$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = mr^2\dot{\theta} = rp_\theta = L = constant$$

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Effective potential:

- If we want to solve a two-dimensional problem (which could have come from a three-dimensional problem) involving a central force, then we can recast the problem into a simple one-dimensional problem with a slightly modified potential.
- We can forget that we ever had the variable $\dot{\theta}$, and we can solve this one-dimensional problem to obtain $r(t)$.

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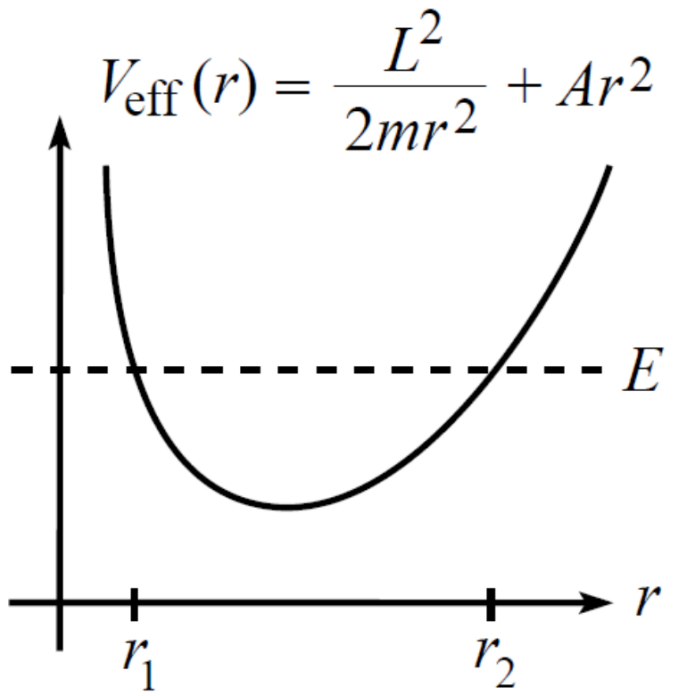
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An example:

$$V(r) = Ar^2$$



CENTRAL FORCE: THE SOLUTION

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Effective potential: A useful method for simplifying a three-dimensional central-force problem down to a one-dimensional problem.

$$V_{\text{eff}}(r) \equiv \frac{L^2}{2mr^2} + V(r) \quad \Rightarrow \quad F_{\text{eff}}(r) = \frac{L^2}{mr^3} - V'(r)$$

CENTRAL FORCE: THE SOLUTION

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To get an actual $r(t)$ out of this, we must be supplied with E and L (determined by the initial values of r, $\dot{r}$, and $\dot{\theta}$), and also the function $V(r)$.

CENTRAL FORCE: THE SOLUTION

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$$r(t) \quad \Rightarrow \quad \dot{\theta}(t) = L/mr^2$$

CENTRAL FORCE: THE SOLUTION

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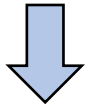
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Finding $r(\theta)$



$$\left(\frac{1}{r^2}\frac{dr}{d\theta}\right)^2 = \frac{2mE}{L^2} - \frac{1}{r^2} - \frac{2mV(r)}{L^2}$$

Followed by separation of variables & integration

CENTRAL FORCE: GRAVITY

Finding $r(\theta)$

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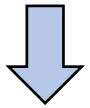
$$V(r) = -\frac{\alpha}{r}, \quad \text{where } \alpha \equiv GM_{\odot}m$$

CENTRAL FORCE: GRAVITY

Finding $r(\theta)$

$$\left(\frac{1}{r^2}\frac{dr}{d\theta}\right)^2 = \frac{2mE}{L^2} - \frac{1}{r^2} - \frac{2mV(r)}{L^2}$$

$$V(r) = -\frac{\alpha}{r}, \quad \text{where } \alpha \equiv GM_{\odot}m$$



$$\frac{1}{r} = \frac{m\alpha}{L^2}(1 + \epsilon\cos\theta),$$

$$\epsilon \equiv \sqrt{1 + \frac{2EL^2}{m\alpha^2}}$$

CENTRAL FORCE: GRAVITY

Finding $r(\theta)$

$$\left(\frac{1}{r^2}\frac{dr}{d\theta}\right)^2 = \frac{2mE}{L^2} - \frac{1}{r^2} - \frac{2mV(r)}{L^2}$$

$$V(r) = -\frac{\alpha}{r}, \quad \text{where } \alpha \equiv GM_{\odot}m$$

$$\frac{1}{r} = \frac{m\alpha}{L^2}(1 + \epsilon\cos\theta),$$

$$r_{\min} = \frac{L^2}{m\alpha(1+\epsilon)}$$

$$r_{\max} = \frac{L^2}{m\alpha(1-\epsilon)} \quad (\text{if } \epsilon < 1)$$

$$\epsilon \equiv \sqrt{1 + \frac{2EL^2}{m\alpha^2}}$$

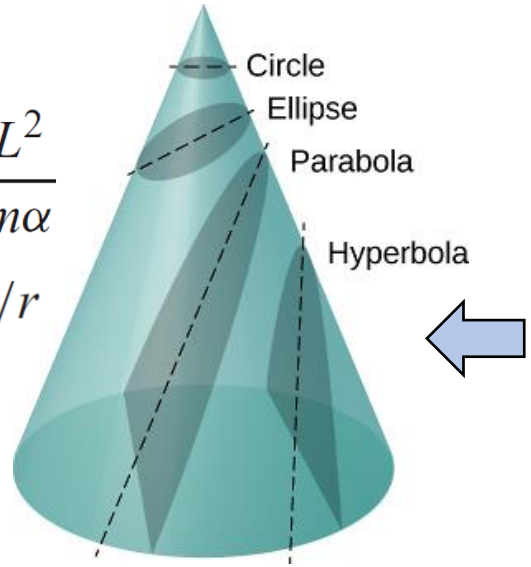
$$r_{\max} = \infty \quad (\text{if } \epsilon \geq 1)$$

CENTRAL FORCE: GRAVITY

Finding $r(\theta)$

$$\left(\frac{1}{r^2}\frac{dr}{d\theta}\right)^2 = \frac{2mE}{L^2} - \frac{1}{r^2} - \frac{2mV(r)}{L^2}$$

$$V(r) = -\frac{\alpha}{r}, \quad \text{where } \alpha \equiv GM_\odot m$$



$$k \equiv \frac{L^2}{m\alpha}$$

$$\cos\theta = x/r$$

$$\frac{1}{r} = \frac{m\alpha}{L^2}(1 + \epsilon\cos\theta),$$

$$\epsilon \equiv \sqrt{1 + \frac{2EL^2}{m\alpha^2}}$$

$$r_{\min} = \frac{L^2}{m\alpha(1+\epsilon)}$$

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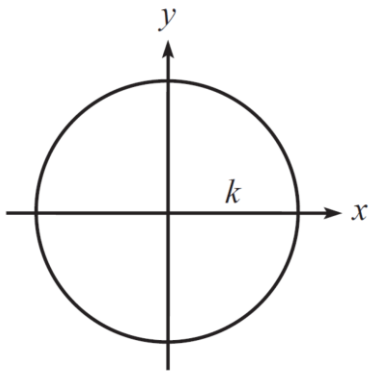
$$x^2 + y^2 = k^2 - 2k\epsilon x + \epsilon^2 x^2$$

CENTRAL FORCE: GRAVITY

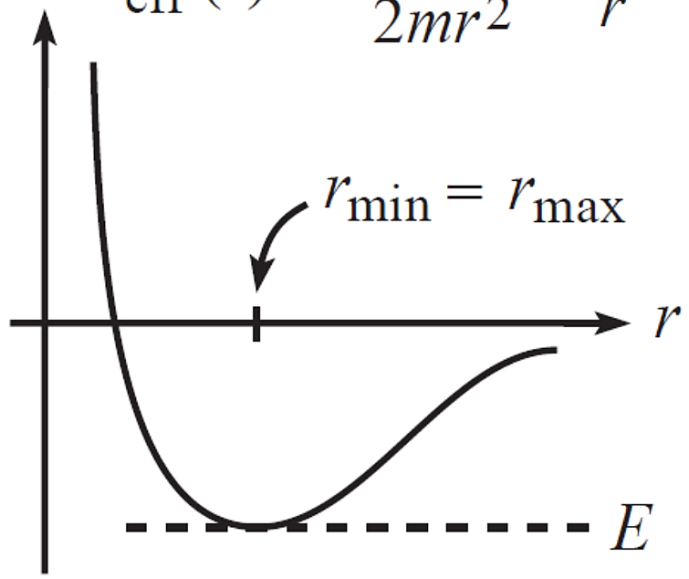
- **Circle** ($\epsilon = 0$)

$$\frac{1}{r} = \frac{m\alpha}{L^2}(1 + \epsilon \cos\theta), \quad \Rightarrow \quad x^2 + y^2 = k^2$$

$$\epsilon \equiv \sqrt{1 + \frac{2EL^2}{m\alpha^2}} \quad \Rightarrow \quad E = -m\alpha^2/2L^2$$



$$V_{\text{eff}}(r) = \frac{L^2}{2mr^2} - \frac{\alpha}{r}$$



$$r_{\min} = r_{\max} = L^2/m\alpha$$

CENTRAL FORCE: GRAVITY

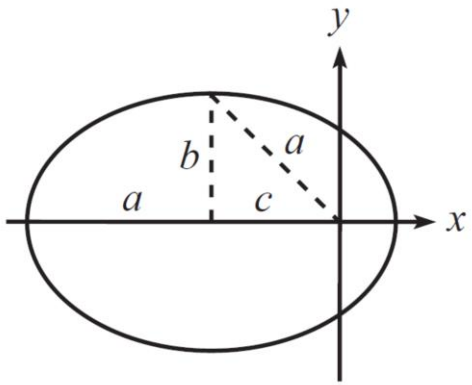
- **Ellipse** $(0 < \epsilon < 1)$

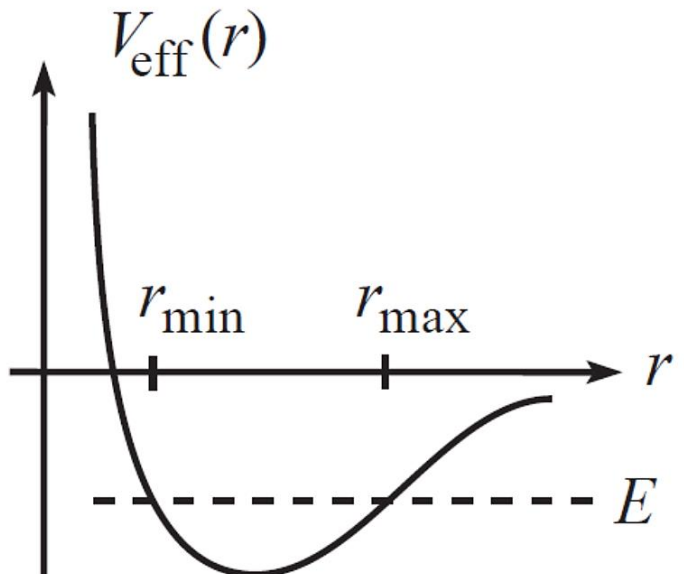
$$\frac{1}{r} = \frac{m\alpha}{L^2}(1 + \epsilon\cos\theta),$$

$$\frac{\left(x + \frac{k\epsilon}{1-\epsilon^2}\right)^2}{a^2} + \frac{y^2}{b^2} = 1,$$

$$a = \frac{k}{1-\epsilon^2}, \quad \text{and} \quad b = \frac{k}{\sqrt{1-\epsilon^2}}$$

$$\epsilon \equiv \sqrt{1 + \frac{2EL^2}{m\alpha^2}} \quad \Rightarrow \quad -m\alpha^2/2L^2 < E < 0$$





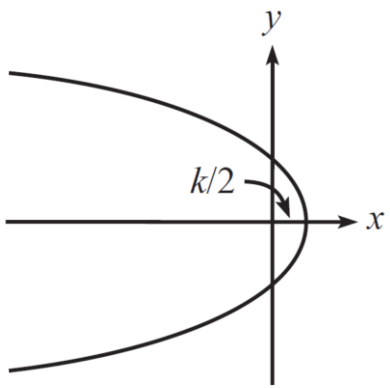
$$r_{\text{min}} = \frac{L^2}{m\alpha(1+\epsilon)} \qquad r_{\text{max}} = \frac{L^2}{m\alpha(1-\epsilon)}$$

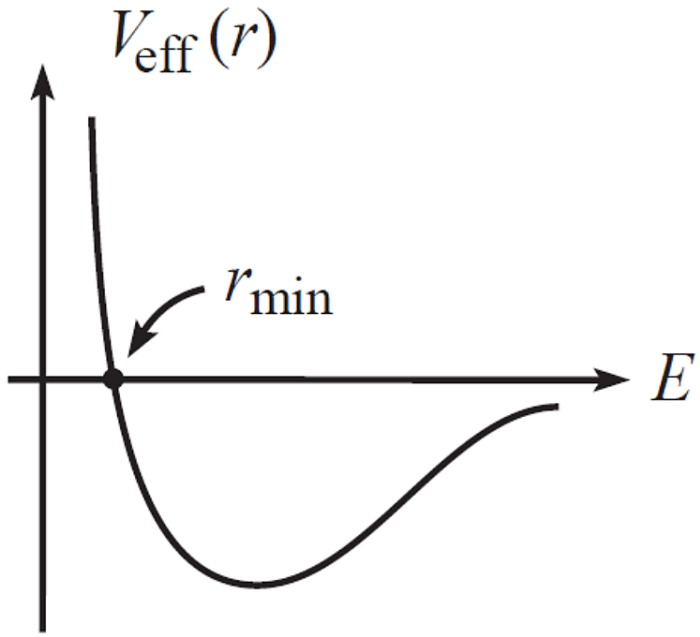
CENTRAL FORCE: GRAVITY

- **Parabola** ($\epsilon = 1$)

$$\frac{1}{r} = \frac{m\alpha}{L^2}(1 + \epsilon\cos\theta), \quad \Rightarrow \quad y^2 = k^2 - 2kx$$

$$\epsilon \equiv \sqrt{1 + \frac{2EL^2}{m\alpha^2}} \quad \Rightarrow \quad E = 0$$





$$r_{\text{min}} = L^2/2m\alpha \qquad r_{\text{max}} = \infty$$

This value of E implies that the particle barely makes it out to infinity (its speed approaches zero as $r \rightarrow \infty$)

CENTRAL FORCE: GRAVITY

- **Hyperbola** ($\epsilon > 1$)

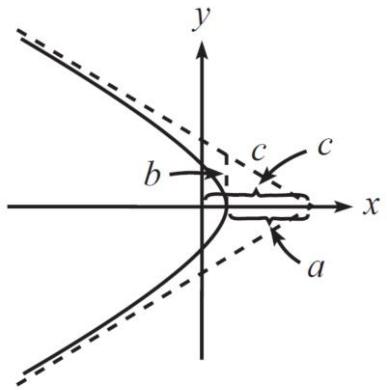
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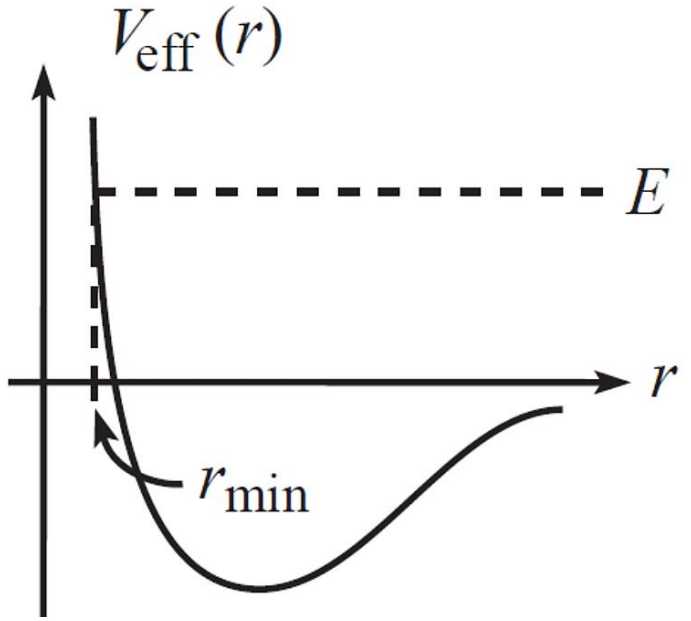
$$\frac{\left(x - \frac{k\epsilon}{\epsilon^2-1}\right)^2}{a^2} - \frac{y^2}{b^2} = 1,$$

$$a = \frac{k}{\epsilon^2 - 1}, \quad \text{and} \quad b = \frac{k}{\sqrt{\epsilon^2 - 1}}$$

$$\epsilon \equiv \sqrt{1 + \frac{2EL^2}{m\alpha^2}}$$

$$E > 0$$





$$r_{\text{min}} = \frac{L^2}{m\alpha(1+\epsilon)} \qquad r_{\text{max}} = \infty$$